



Tausi

Area Review – Chloris 30m
October 2025

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Request: Spatial assessment on project areas

- Review forest/non forest cover over the AOI for 2007-2024

Summary of Analysis

- Chloris 30m data shows >99% forest cover in the AOI from 2007 onwards
- Forest cover appears stable over time based on this classification
- However, visual comparison with satellite imagery. ESA Landcover and Canopy height data, suggests some areas may have more non-forest than the data indicates
- This raises questions about the sensitivity of binary forest/non-forest classification for detecting subtle changes
- Above Ground Biomass data might offer a more nuanced view of vegetation condition and change over time
 - Example using 250 t CO₂ eq/ha as threshold (>slide 9)
 - Forest area appears to increase since 2007
 - Threshold and approach need review (Other threshold results in other result)
 - Questions:
 - Is 250 t CO₂ eq/ha the right threshold?
 - Use binary or multi-class classification?
 - Example plots of planted areas?
- If we agree on the classification, continuous forest areas can be extracted from the data for further analysis



1

Hansen

Subtitle/additional information if required

Forest/Non Forest (Hansen 30m)

Based on Hansen Global Forest Change data:

- Over the past years, the forest cover in the area is stable.
- Dataset limitations in Uganda:
 - Less accurate in dry forests and agroforestry landscapes
 - May underestimate forest cover in mosaic or community-managed areas

Use Hansen data with caution. Higher-resolution data from Chloris is available, but only partially covers the area and was therefore not used in this analysis.

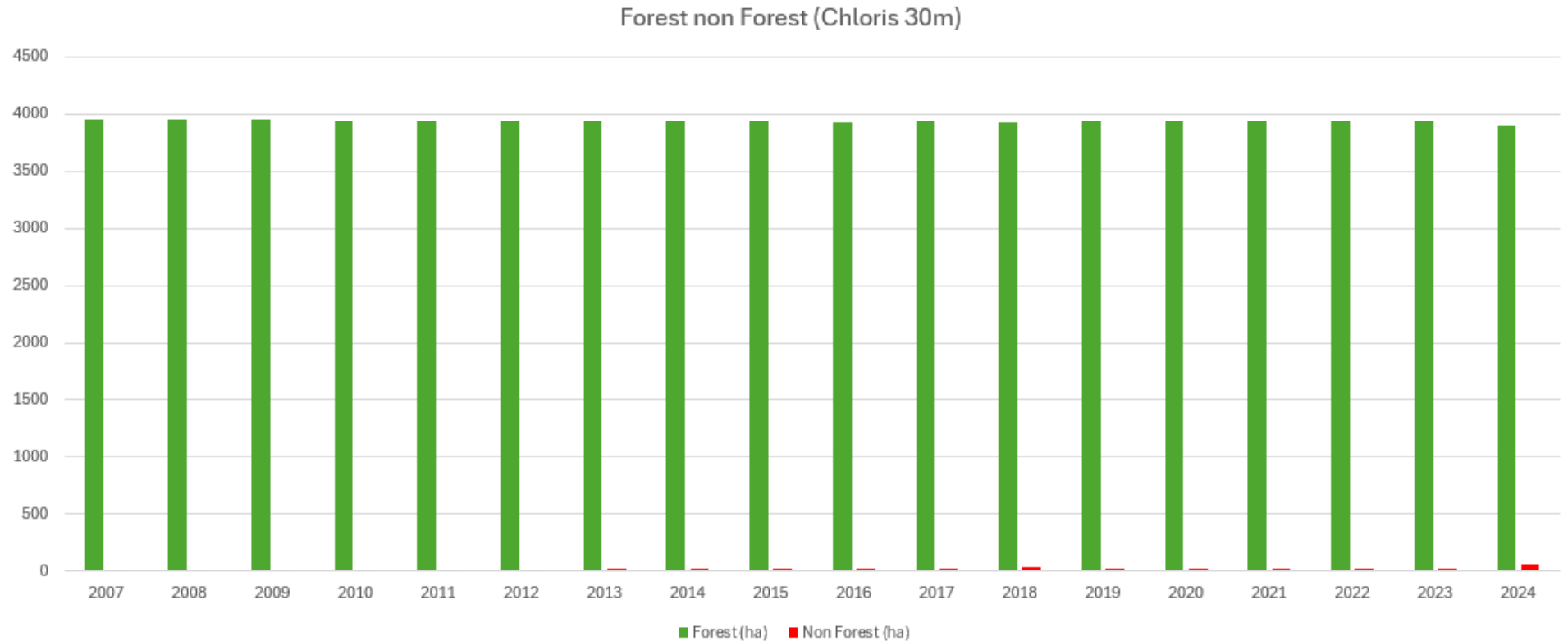
2

Chloris

Subtitle/additional information if required

Forest/Non Forest (Chloris 30m)

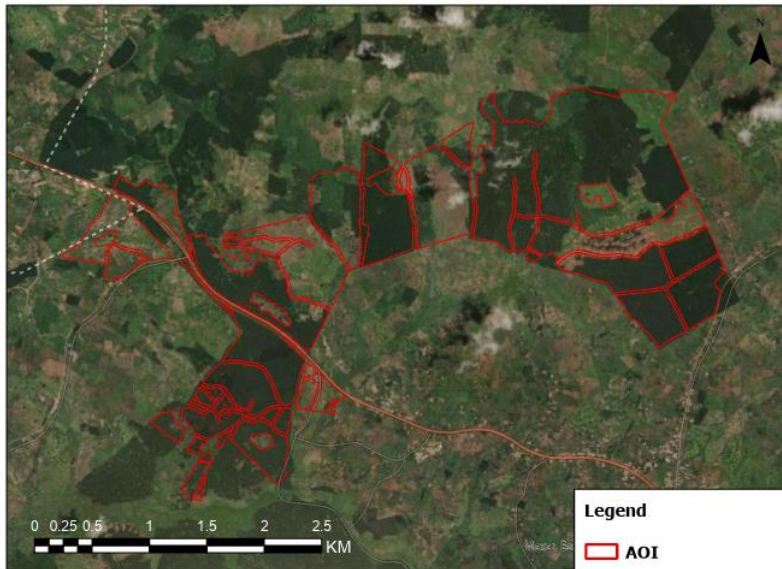
- Chloris 30m data was used to assess forest and non-forest areas
- Results show that over 99% of the AOI has been classified as forest from 2007 onwards
- Forest cover has remained stable across the AOI since 2007



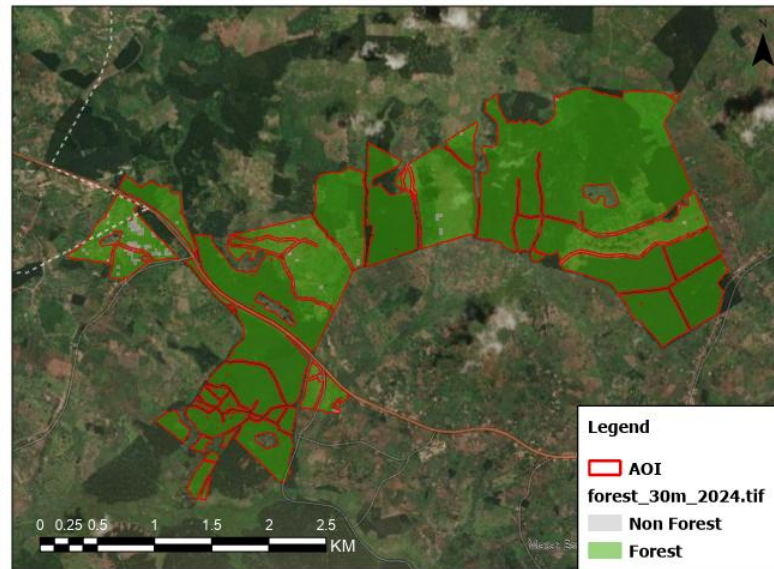
Forest vs. Non-Forest vs. Above Ground Biomass (Chloris 30m)

- Chloris data shows 99% forest cover, but imagery suggests discrepancies.
- Carbon stock trends may provide a better indicator.

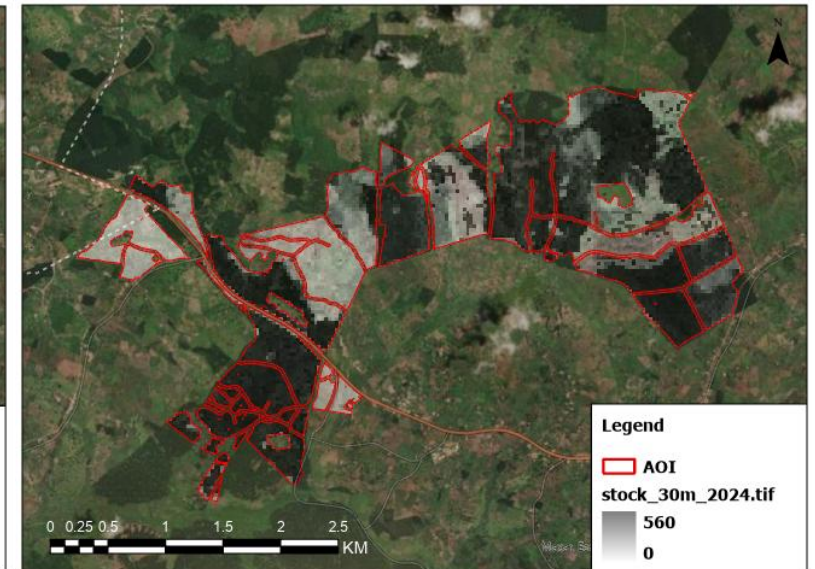
Imagery



Forest/Non-Forest

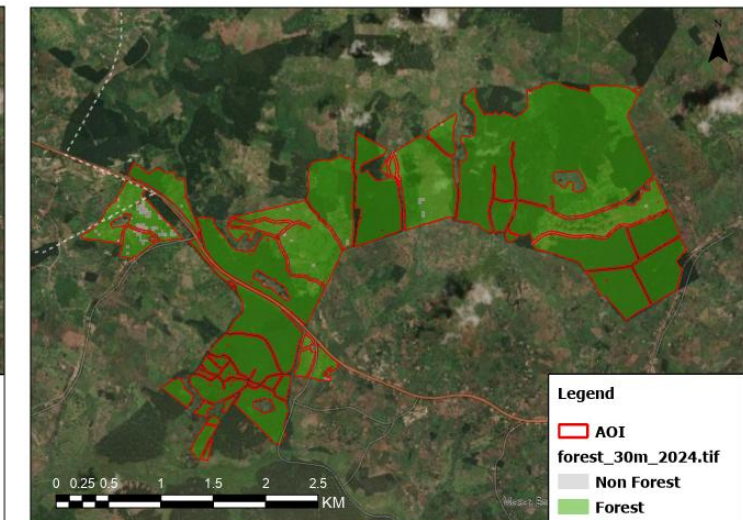
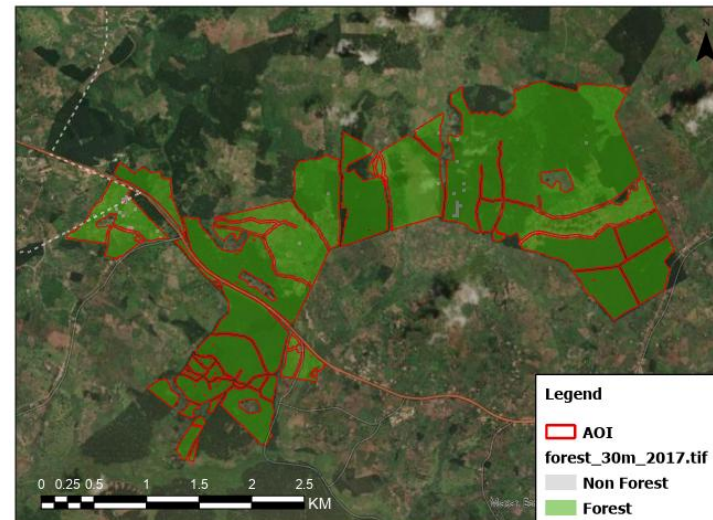
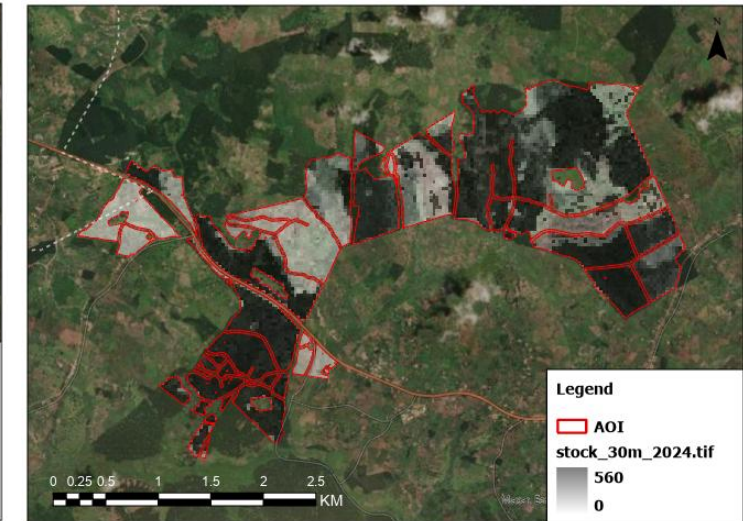
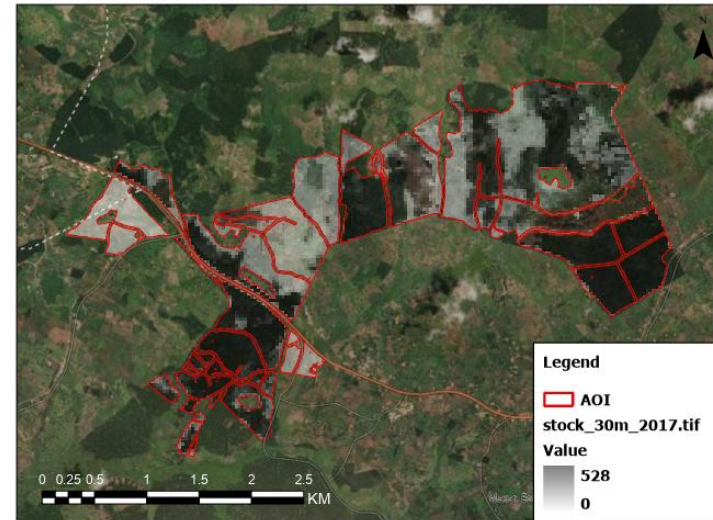


Above Ground Biomass



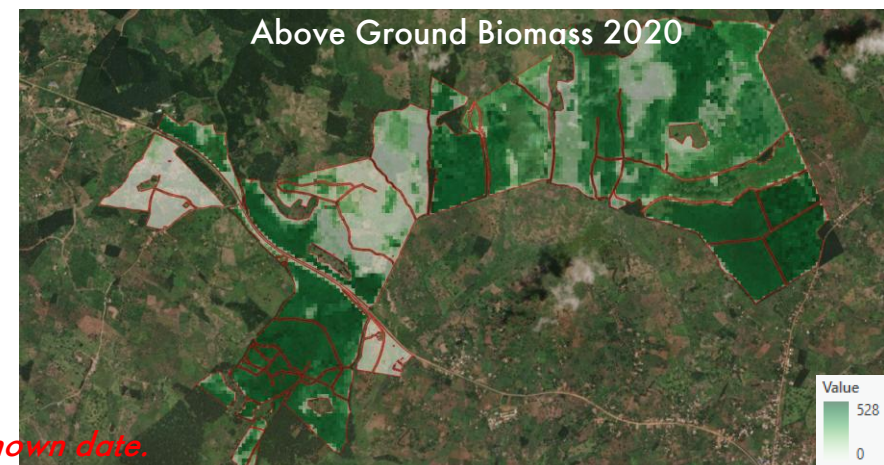
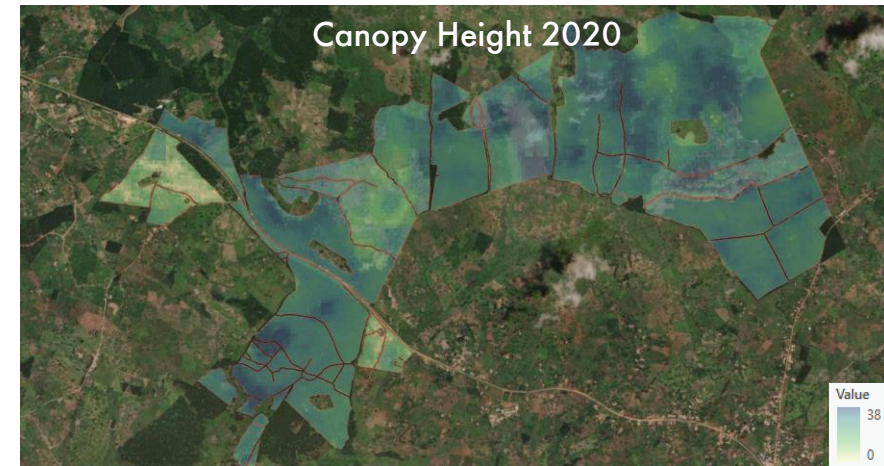
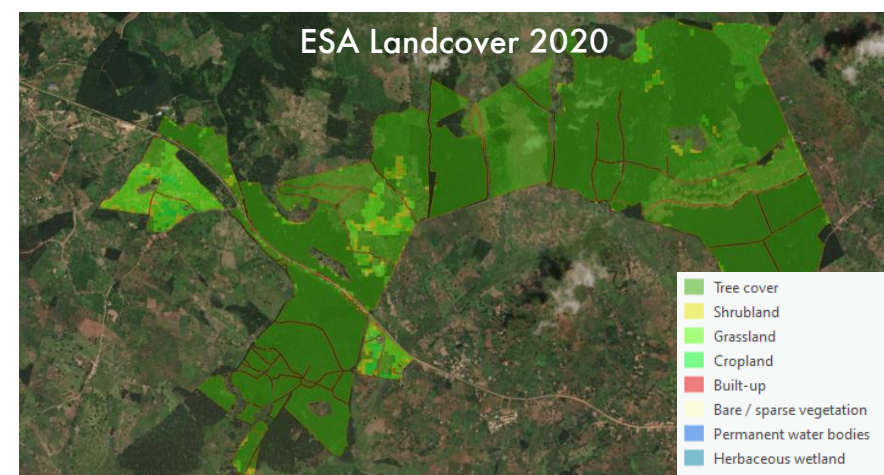
Above Ground Biomass (Chloris 30m)

- Reviewing AGB data over time instead of relying solely on forest/non-forest classification
- Provides a more nuanced view of vegetation changes, especially in areas where satellite imagery suggests discrepancies
- Classification of the AGB time series could help identify subtle trends or degradation not captured by binary forest classification.
- AGB is ton co2 eq per ha



AGB & Reference Layers (2020)

- ESA land cover shows mostly tree cover
- Canopy height and AGB patterns align
- Using AGB classified with ESA and canopy height as reference layers can improve Forest/Non-Forest mapping.
- Satellite imagery underlay is useful for context, but acquisition date is unknown
- Next step: Test Classification of AGB with ESA and canopy height to refine Forest/Non-Forest mapping

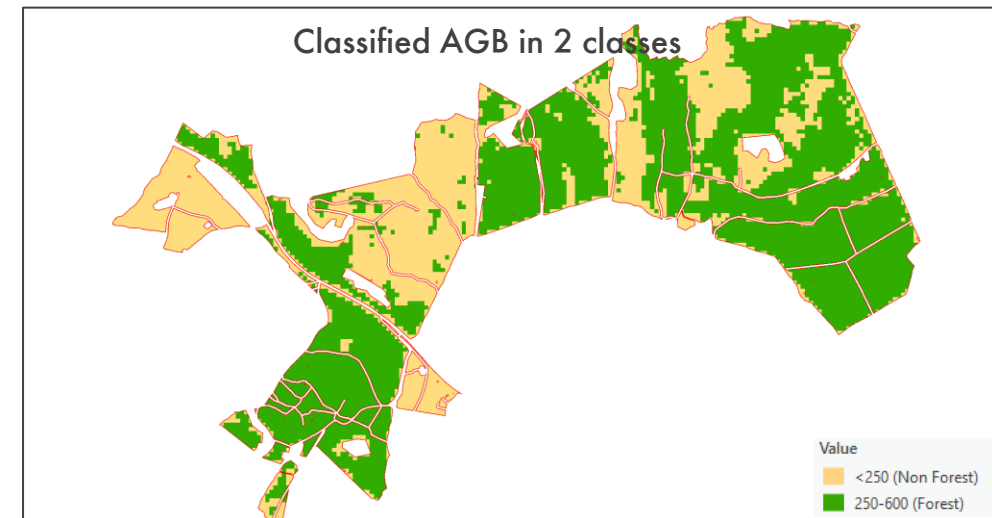
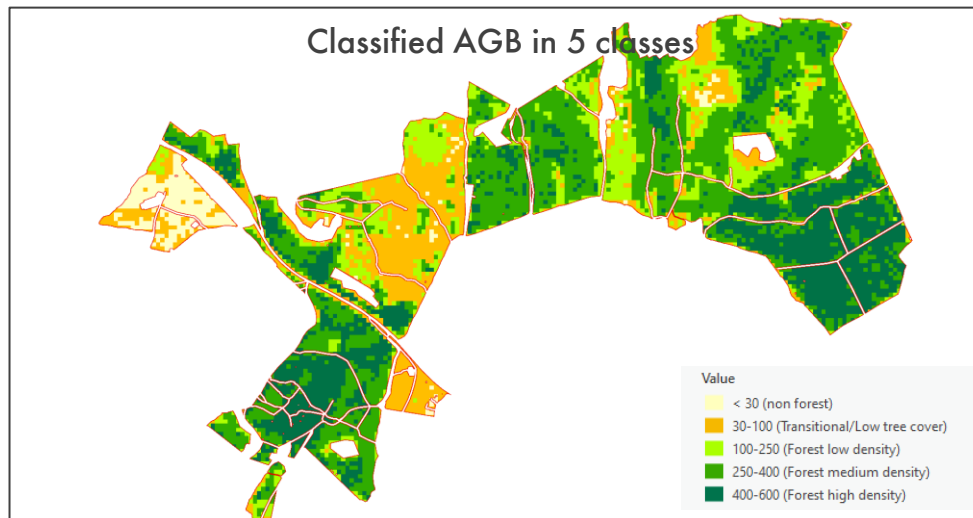
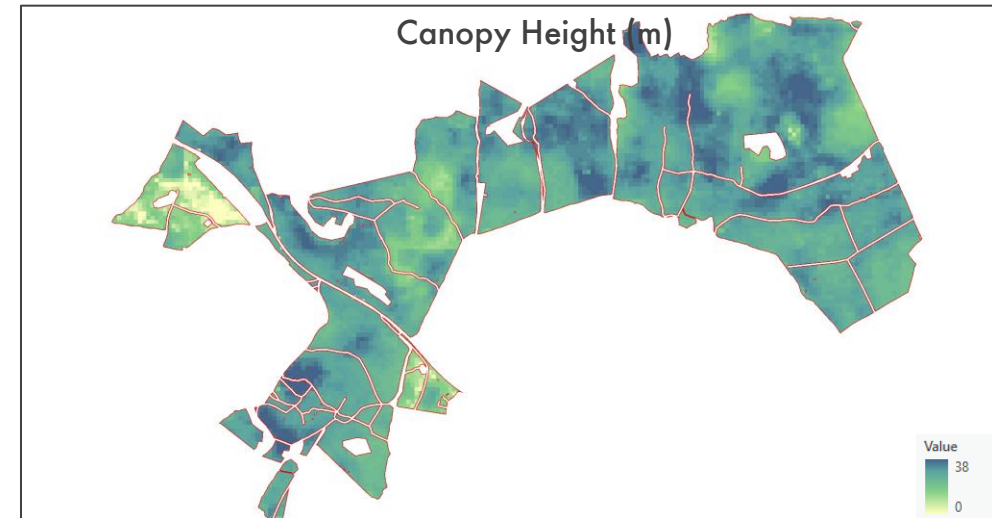
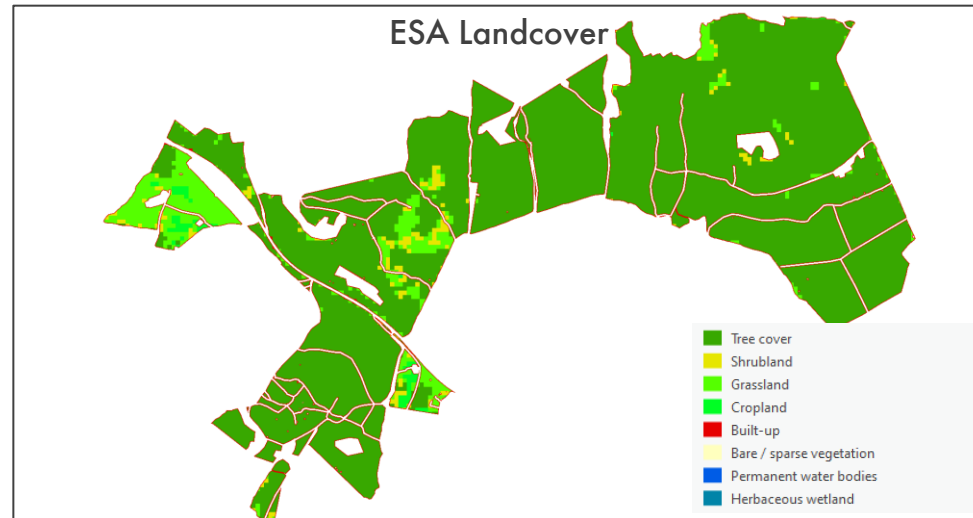


Classifying Above-Ground Biomass for Forest/Non-Forest

- Overlaying ESA land cover and canopy height with AGB data (2020) suggests the following classification thresholds, based on literature and internal analysis.
- Example Simplified rule:
 - Forest > 250 t CO₂ eq/ha
 - Non-Forest ≤ 250 t CO₂ eq/ha

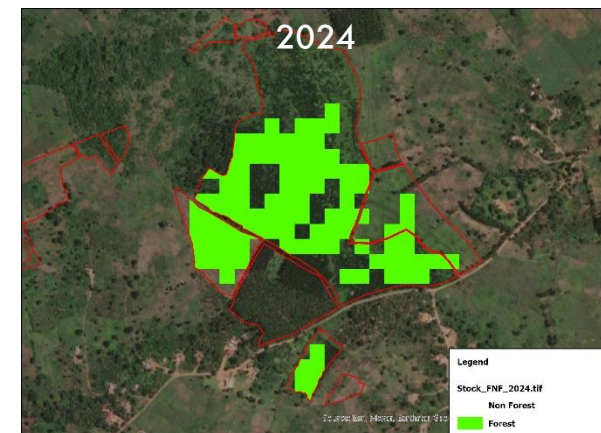
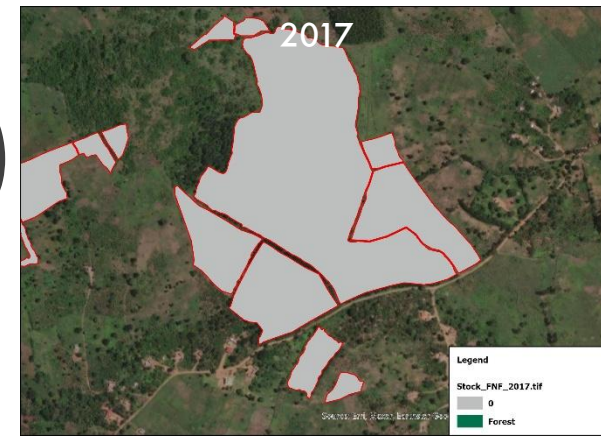
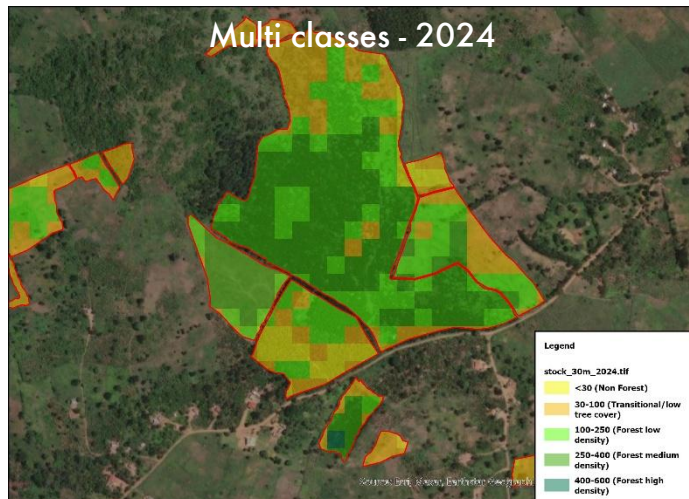
Biomass (ton CO2 eq /ha)	Interpretation	Class
0-30	Bare / grass / cropland	Non forest
30-100	Shrubland / sparse trees	Transitional / Low Tree Cover
100-250	Open woodland / young plantations	Forest (Low Density)
250-400	Mature woodland / mixed forest	Forest (Medium Density)
400-600+	Dense forest / plantations	Forest (High Density)

AGB & Reference Layers (2020)



Binary AGB Classification: Planted Area (Example)

- Example using 250 t CO₂e/ha as threshold.
- Binary classification suggests forest growth from 2017 to 2024.
- Multi-class view indicates the threshold may be too high.
- If we agree on classification, continuous forest areas can be extracted for further analysis.



The background in all images is satellite imagery without known date.

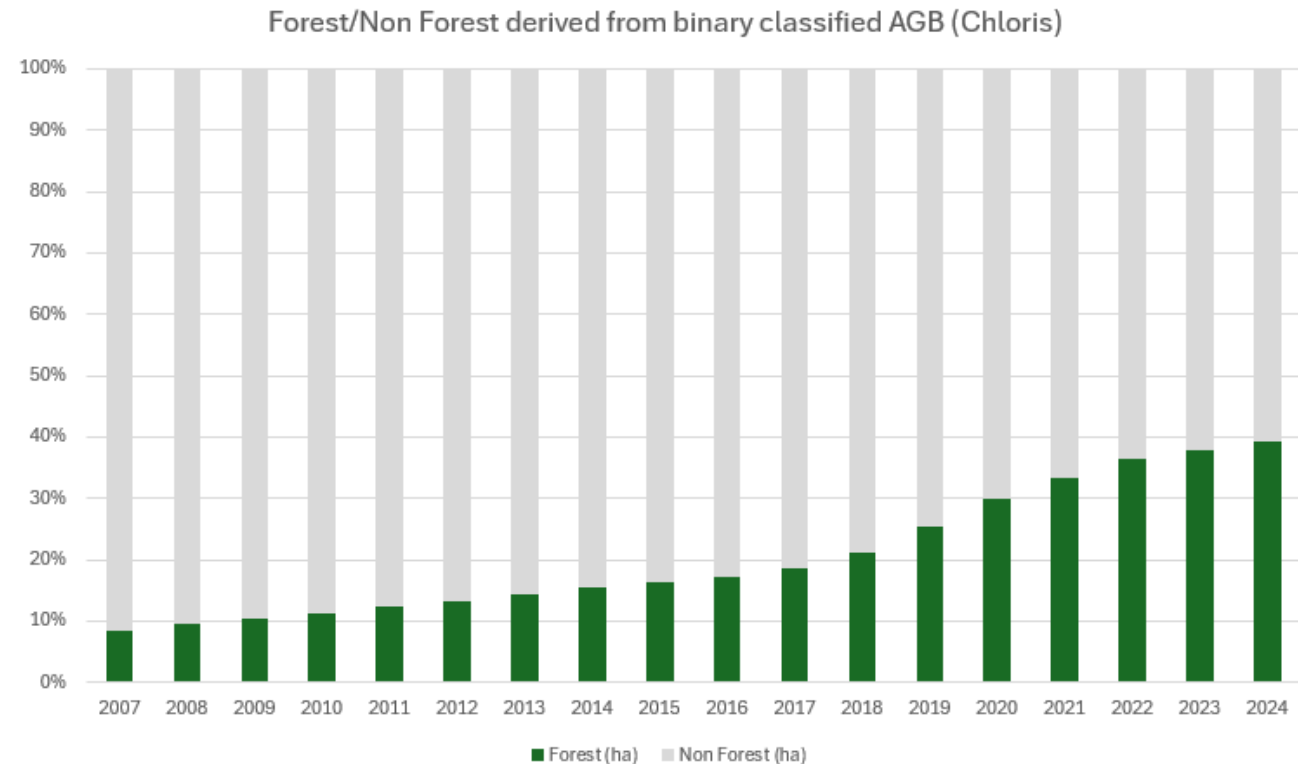
INTERNAL

Binary Classification of AGB: Forest vs. Non-Forest (Example)

- Example using 250 t CO₂ eq/ha as threshold
- Forest area appears to increase since 2007
- Threshold and approach need review
- Other threshold results in other result

- Questions:

- Is 250 t CO₂ eq/ha the right threshold?
 - Use binary or multi-class classification?
 - Example plots of planted areas?
- If we agree on the classification, continuous forest areas can be extracted from the data for further analysis*



2011

2016

2021

Legend

stork_30m_2011.tif

<250 (Non Forest)

250-600 (Forest)

Legend

stork_30m_2016.tif

<250 (Non Forest)

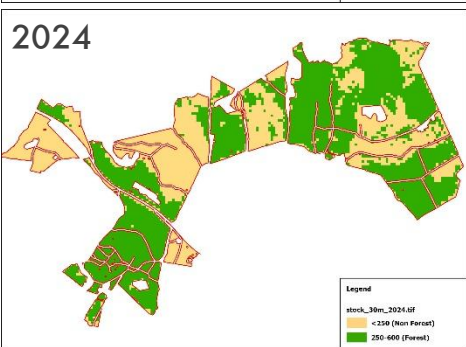
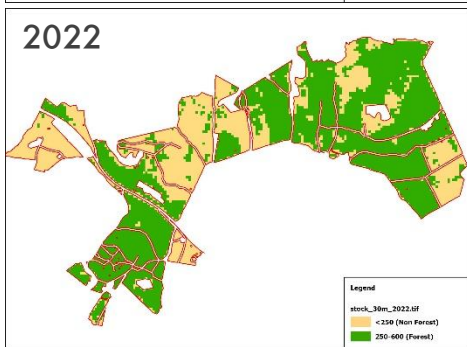
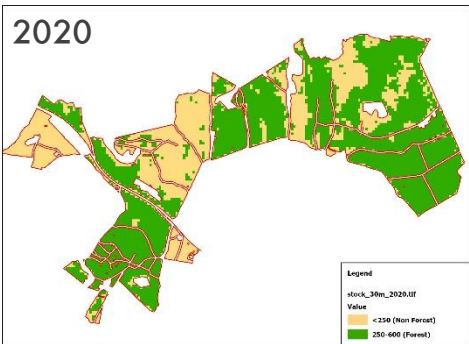
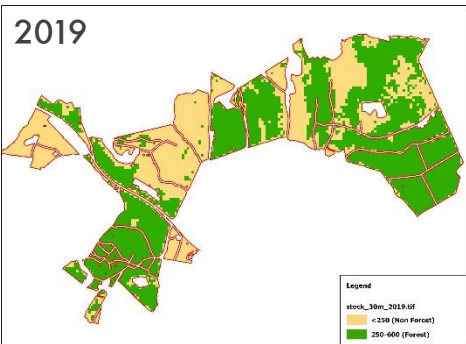
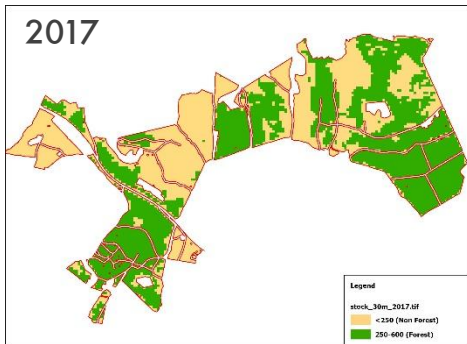
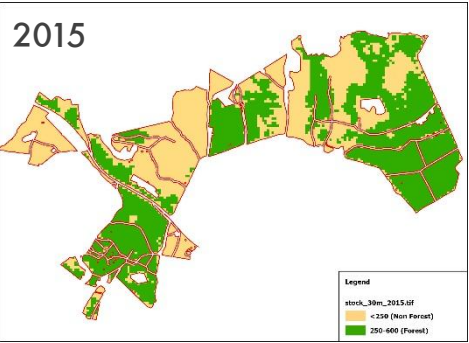
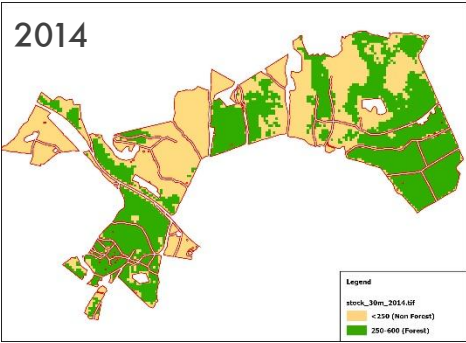
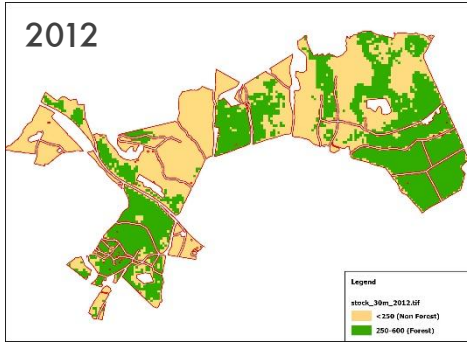
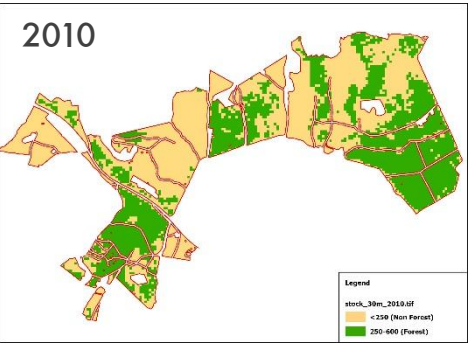
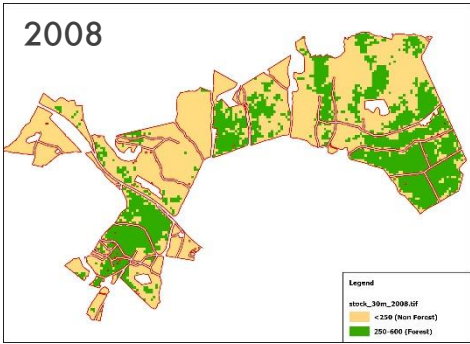
250-600 (Forest)

Legend

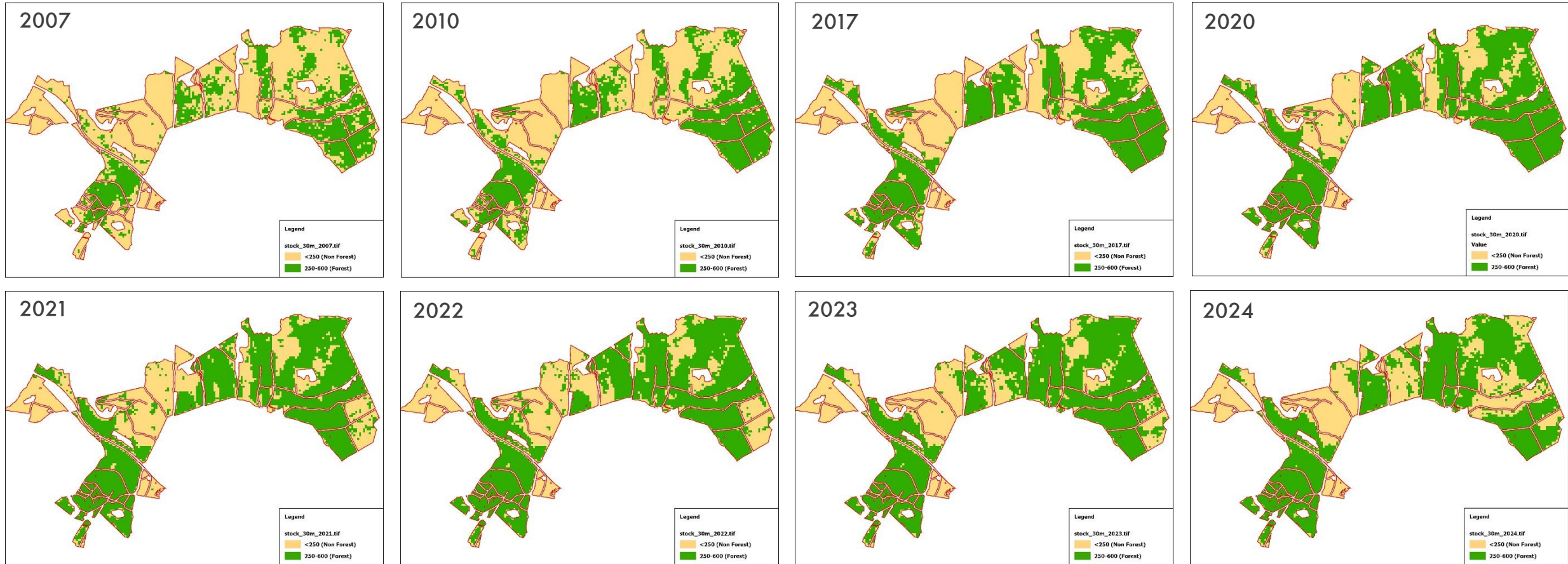
stork_30m_2021.tif

<250 (Non Forest)

250-600 (Forest)



AGB 2007, 2010, 2017, 2020-2024



3

Landcover

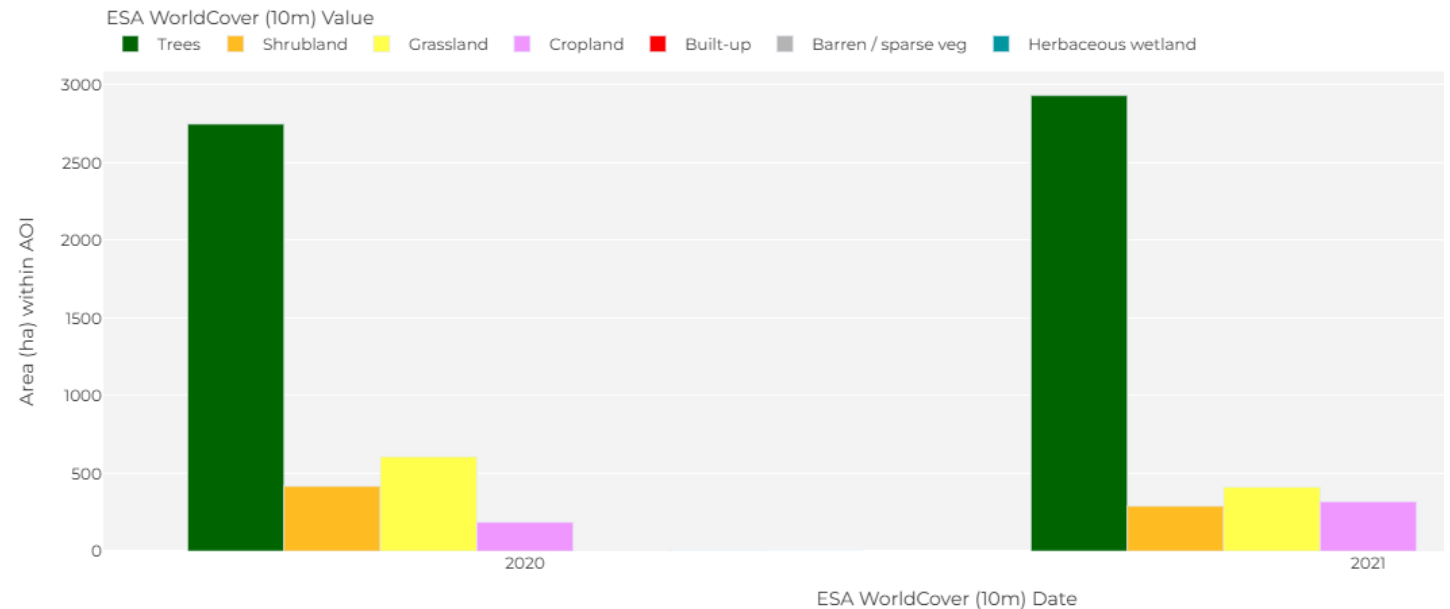
Subtitle/additional information if required

ESA Landcover (10m)

- ESA 10m data was used to assess the landcover in 2020 and 2021
- ESA WorldCover is trustworthy for Uganda
 - Datasource: Sentinel data
 - Validated accuracy for Africa
 - Consistent classification scheme
- Results show that
 - ~68% of AOI classified as forest in 2020
 - ~73% of AOI classified as forest in 2021

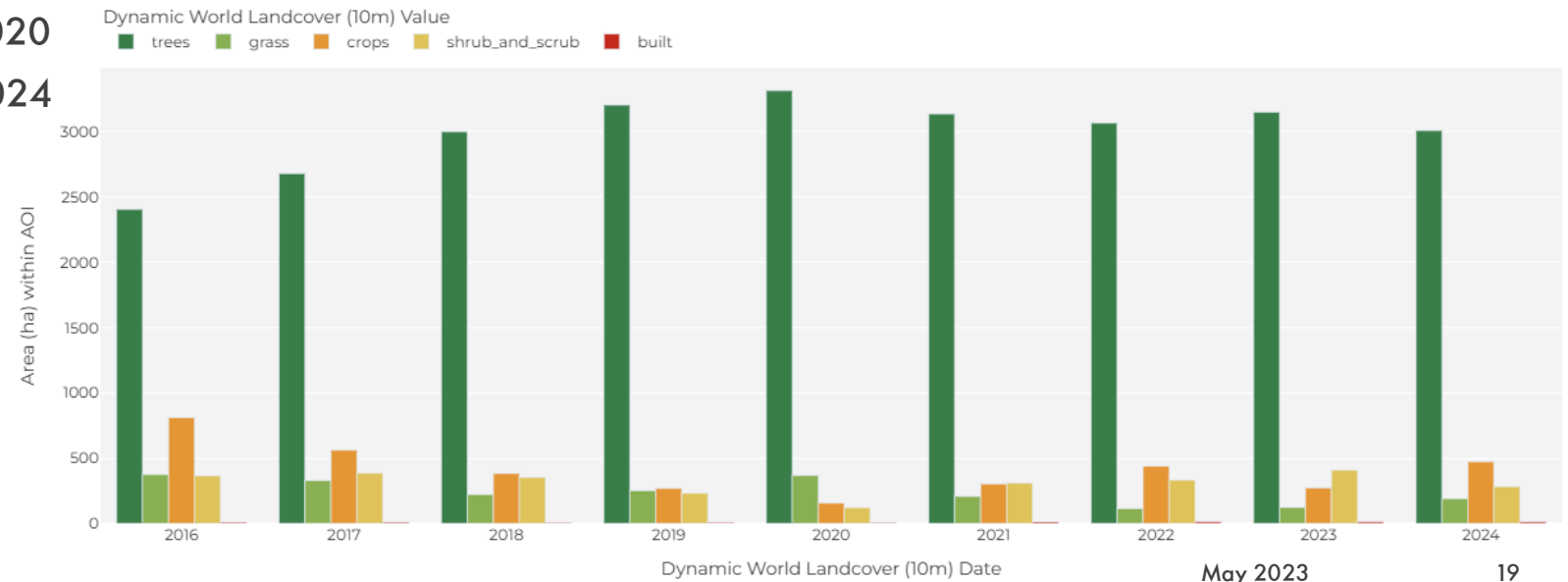
Esa - Landcover

Area (hectares) of each land cover type within the AOI, per year. ESA



Dynamic World Landcover (10m)

- Dynamic World 10m data was used to assess the landcover between 2016 and 2024
- Dynamic World is trustworthy for Uganda
 - Data source: Sentinel data + deep learning
 - Accuracy for Africa ~70%
 - Consistent classification scheme
- Results show that
 - ~60% of AOI classified as forest in 2016
 - ~83% of AOI classified as forest in 2020
 - ~75% of AOI classified as forest in 2024





4

NFC Non-Forest Report

Subtitle/additional information if required

NFC Non-Forest Report

- This report outlines a multi-temporal mapping approach for forest/non-forest classification across Uganda's New Forest (NF) areas of interest (AOI), covering the years 2000, 2005, 2010, and 2020.
- Combination of Sentinel-2, Landsat 5/7, and Global Forest Watch (GFW) datasets.
 - 2020: Derived from Sentinel-2 imagery using NDVI thresholds
 - 2010: Based on the GFW 2010 tree cover dataset, recoded into density classes (0–39%, 40–49%, etc)
 - 2005: NDVI composites from Landsat 7 across multiple quarters, filtered for cloud-free scenes.
 - 2000: GFW 2000 tree cover dataset, resampled to 20 m and recoded into density classes.
- Reprojected to UTM 36N WGS84, aligned to a 20 m grid, and zonal statistics were calculated for each NF compartment polygon.
- Tree cover was classified into qualitative density classes (low, medium, high) and linked to plantation age data.
- The report emphasizes that GFW datasets do not distinguish between indigenous and planted trees, and NDVI-based methods were used to refine interpretations. The authors suggest that forest loss between 2000 and 2010 likely reflects anthropogenic clearing, with some areas later replanted.

NFC Non-Forest Report - Summary

- A direct visual comparison of the 2000 and 2010 GFW-derived tree cover maps shows a significant reduction in tree cover across the NF mapping AOI. This reduction is visible as:
 - Less green class coverage
 - Lower tree cover density values in areas that still have trees in 2010
- Likely due to anthropogenic clearing of indigenous tree and bush cover.
 - Some polygons may show: High tree cover in 2000
 - Reduced or lost tree cover in 2005/2010
 - Reappearance of tree cover in 2020, likely due to plantation replanting
- However, since the GFW datasets do not distinguish between indigenous and planted trees, this cannot be confirmed without a historical land-cover dataset that separates these classes.
- The 2005 NDVI-based tree cover resembles the 2000 map more than the 2010 map, suggesting that most clearing occurred between 2005 and 2010.

NFC Non-Forest Report – My Summary

The approach is multi-temporal with creative combination of datasets over the years to track change

- NDVI-based classification: First pass is good; should be combined with other classifications
- Use of GFW/Hansen datasets: Globally recognized but use with care over Uganda.

Limitations

- No distinction between indigenous vs planted trees: GFW/Hansen datasets treat all tree cover equally.
- No thematic land-use separation for 2000/2010: So you can't confirm whether tree cover is natural forest or plantation.
- For demonstrating that areas were not forested in 2007, the report provides strong indicative evidence, but not absolute proof.
- It supports the argument that large-scale clearing happened before 2010, so plantations after 2020 are likely compliant with the 10-year rule.
- For 2017–2019 plantings, the evidence is weaker because:
 - The report does not provide fine-grained verification for those specific polygons.
 - It cannot confirm no clearing occurred between 2007 and 2010 for each outgrower plot.

NFC Non-Forest Report - 2020

- 2020 Annualised (Plantation Only) Tree / Non-Tree Derived from the 2020 annualised land-cover dataset
 - Focuses only on standing plantation timber areas.
 - Excludes indigenous woody cover and temporarily clear-felled compartments.
 - Wetlands from the 2020 base dataset were included to avoid misclassification.
 - Zonal statistics were calculated for each NF polygon
- They acknowledge that GFW may misclassify non-tree vegetation due to spectral similarity.

NFC Non-Forest Report - 2010

- The 2010 tree/non-tree map was derived from the Global Forest Watch (GFW) 2010 tree cover dataset
 - This dataset provides percent tree cover (0–100%) per pixel, without distinguishing between indigenous vs planted trees.
 - Due to lack of usable cloud-free Landsat imagery in 2010, no NDVI-based quarterly composites could be created for that year.
 - The GFW dataset was recoded into density classes: 0–39%, 40–49%, 50–59%, 60–74%, and 75–100%
 - Wetlands were masked out using the 2020 land cover dataset to avoid misclassification of high-biomass non-tree areas.
 - The GFW data was resampled from 30 m to 20 m using nearest-neighbour interpolation to match the resolution of the 2020 dataset.
 - Zonal statistics were calculated for each NF compartment and outgrower polygons
-
- The document confirms that the entire AOI was classified as forest in the GFW dataset
 - The authors used tree cover density classes instead of binary forest/non-forest to better reflect variation.
 - They acknowledge that GFW may misclassify non-tree vegetation due to spectral similarity.

NFC Non-Forest Report - 2005

- The 2005 tree/non-tree map was created using Landsat 7 imagery from 2004 Q2, 2004 Q4, 2005 Q1–Q4, and 2006 Q1
- GFW-derived tree cover data from 2000 and 2010 for reference
- For each quarter, the maximum NDVI value per pixel was extracted to reduce cloud impact.
- These quarterly NDVI layers were combined into a single annualised NDVI composite for 2005.
- Cloudy quarters were excluded from the final composite.
- NDVI thresholds were manually defined per quarter using visual reference areas to separate woody vs non-woody cover.
- The final 2005 tree cover map was spatially constrained to match areas with $\geq 40\%$ tree cover in both the 2000 and 2010 GFW datasets.
- Wetlands were masked using the 2020 base land-cover dataset to avoid misclassification.
- The final raster was resampled to 20 m resolution (from 30 m) using nearest-neighbour interpolation.
- Zonal statistics were calculated for each NF polygon

NFC Non-Forest Report - 2000

- The 2000 tree/non-tree map was derived from the Global Forest Watch (GFW) 2000 tree cover dataset
 - Specifically the file Hansen_GFC-2020-v1.8_treecover2000_10N_030.tif.
- This dataset provides percent tree cover (0–100%) per pixel, without distinguishing between indigenous vs planted trees.
- The data was recoded into density classes: 0–39%, 40–49%, 50–59%, 60–74%, and 75–100%
- Wetlands were masked using the 2020 base land-cover dataset to avoid misclassification of high-biomass non-tree areas.
- The raster was resampled from 30 m to 20 m using nearest-neighbour interpolation to match the resolution of the 2020 dataset.
- Zonal statistics were calculated for each NF compartment and outgrower polygon



INTERNAL